

Regression Analysis

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Outline

Correlation

Regression

Lab Report

Covariance

The **Covariance** between two variables can be estimated by:

$$\text{cov}(X, Y) = s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Code

```
cov(fert$Fertiliser, fert$Yield)
cov(fert$Yield, fert$Fertiliser)
```

Output

```
> cov(fert$Fertiliser, fert$Yield)
[1] 25.54167
> cov(fert$Yield, fert$Fertiliser)
[1] 25.54167
```

Correlation Coefficient

Pearson's correlation measure standardizes the covariance by the variances of the original variables, so that its value always falls between **-1 and +1**. It is defined as:

$$r_{XY} = \frac{S_{XY}}{\sqrt{S_X^2 S_Y^2}} = \frac{S_{XY}}{S_X S_Y}$$

Code

```
cor(fert$Fertiliser, fert$Yield)
```

Output

```
> cor(fert$Fertiliser, fert$Yield)
[1] 0.8559683
```

Test of significance for correlation

A method that tests whether a sample correlation coefficient (r) could have come from a population with **zero correlation** ($\rho = 0$), i.e., whether the two variables are truly associated.

Hypothesis:

$$H_0 : \rho = 0 \quad (\text{the two variables are uncorrelated})$$

$$H_1 : \rho \neq 0$$

Test Statistic:

$$t_0 = \frac{r - 0}{SE(r)} = r \sqrt{\frac{n - 2}{1 - r^2}}$$

where $SE(r) = \sqrt{\frac{1 - r^2}{n - 2}}$ is the standard error of r , with $df = n - 2$.

Decision Rule:

Reject H_0 if $|t_0| > t_{\alpha/2, n-2}$ (the correlation is significant).

Test of significance for correlation

Code

```
cor.test(fert$Fertiliser, fert$Yield)
```

Output

```
Pearson's product-moment correlation

data: fert$Fertiliser and fert$Yield
t = 4.3802, df = 7, p-value = 0.003234
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.4445700 0.9691564
sample estimates:
      cor 
0.8559683
```

Fitting a linear model in R

The simple linear regression model:

$$\hat{y}_i = a + b x_i$$

where a is the intercept and b is the slope. We use `lm()` to estimate a and b .

Code

```
model <- lm(Mortality ~ Exposure, data = mort)
model
```

Output

```
Call:
lm(formula = Mortality ~ Exposure, data = mort)

Coefficients:
(Intercept)      Exposure
    114.716         9.231
```

Reading the output: coefficients

Code

```
summary(model)
```

Output (coefficients)

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	114.716	8.046	14.258	1.98e-06	***
Exposure	9.231	1.419	6.507	0.000332	***

What each number means (for Exposure)

- ▶ Estimate = 9.231 → slope b
- ▶ Std. Error = 1.419 → $SE(b)$
- ▶ t value = 6.507 → $t_0 = b/SE(b)$
- ▶ Pr(>|t|) = 0.0003 → p -value for $H_0 : \beta = 0$

Coefficient of determination (r^2)

How much of the variability in Y is explained by X ?

$$r^2 = \frac{SS_{Regression}}{SS_{Total}}$$

Definition: the proportion of the total variability in Y that is explained by the regression on X .

- ▶ $r^2 \rightarrow 1$: the model explains most of the variability.
- ▶ $r^2 \rightarrow 0$: the model explains very little.
- ▶ Note: $\sqrt{r^2} = |r|$ (links regression back to correlation).

Adjusted r^2

Why not just use r^2 ? Adding more predictors **always** increases r^2 , even useless ones.

$$r_{adj}^2 = 1 - (1 - r^2) \frac{n - 1}{n - k - 1}$$

Definition: a version of r^2 that is penalised for the number of predictors (k) in the model.

- ▶ It only increases if a new predictor improves the model more than expected by chance.
- ▶ $r_{adj}^2 \leq r^2$ (always equal to or smaller than r^2).
- ▶ Useful for comparing models with different numbers of predictors.

The F-test in regression

The total variation in Y is partitioned into two parts:

$$SS_{Total} = SS_{Regression} + SS_{Residual}$$

The **F-test** compares the *explained* variation against the *unexplained* variation:

$$F_0 = \frac{MS_{Regression}}{MS_{Residual}}$$

Hypothesis:

$$H_0 : \beta = 0 \quad H_1 : \beta \neq 0$$

A large F_0 (small p -value) means X explains significantly more variation than random error $\Rightarrow$ reject H_0 .

Note: For simple linear regression (one X), the F-test and the slope t -test are equivalent: $F_0 = t_0^2 = 6.507^2 = 42.34$, same p -value.

Reading the output: model fit

Output (bottom of summary)

```
Multiple R-squared:  0.8581,   Adjusted R-squared:  0.8378  
F-statistic: 42.34 on 1 and 7 DF,  p-value: 0.0003321
```

What each number means

- ▶ $R\text{-squared} = 0.858 \rightarrow X$ explains 85.8% of the variability in Y
- ▶ $\sqrt{0.858} \approx 0.926 \rightarrow$ the correlation coefficient r
- ▶ $F\text{-statistic} = 42.34, p = 0.0003 \rightarrow$ result of the F-test ($H_0: \beta = 0$)

Conclusion

$F_0 = 42.34, p < 0.05 \Rightarrow$ Reject H_0 .

Exposure explains a significant proportion of the variability in mortality.

Plotting the fitted line

Once we have the model, we can draw the fitted line on the scatter plot using `abline()`.

Code

```
plot(mort$Exposure, mort$Mortality,  
      xlab = "Exposure", ylab = "Mortality")  
abline(model, col = "red")
```

- ▶ `plot()` draws the data points.
- ▶ `abline(model)` adds the regression line directly from the fitted model.

Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 5/31 (Sun) 23:59
- ▶ Late submission is not accepted